

CO1 ASSESSMENT TOOL 1

CASE STUDY ANALYSIS

Online Movie Ticket Booking System

Course Code: CSA1004

Course Title: Software Engineering

Assessment: Assessment Tool 1 – Case Study Analysis

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1. Understand the Problem Statement

1.1 Problem Description

An Online Movie Ticket Booking System is a software application that allows users to search for movies, view theatres and show timings, select seats, make online payments, and receive booking confirmations without physically visiting the theatre.

In the traditional ticket booking process, customers need to visit the theatre or call the theatre to check movie availability and book tickets. This process can be time-consuming, especially during weekends and newly released movies. Customers may also face long queues and may not know which seats are available.

The proposed system provides a convenient online platform where customers can complete the entire ticket-booking process through a website or mobile application.

1.2 Objectives

The main objectives of the system are:

1. To provide online movie ticket booking facilities.
2. To allow users to view currently available movies.
3. To display theatres, show timings, and available seats.
4. To allow users to select their preferred seats.
5. To provide secure online payment facilities.

6. To generate electronic booking confirmation and tickets.
7. To reduce waiting time and manual work.
8. To help theatre administrators manage movies, shows, seats, and bookings efficiently.

1.3 Scope

The scope of the Online Movie Ticket Booking System includes:

- User registration and login.
- Movie search and browsing.
- Theatre selection.
- Show timing selection.
- Real-time seat availability.
- Seat selection and booking.
- Online payment.
- Ticket generation.
- Booking history and cancellation.
- Notifications through email or SMS.
- Theatre administrator management.
- Reports on bookings and revenue.

The system can be implemented as a web application and can later be extended into Android and iOS applications.

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders

1. Customers

Customers use the system to search for movies, select theatres and show timings, choose seats, make payments, and receive tickets.

2. Theatre Administrators

Theatre administrators manage movies, screens, seats, show timings, ticket prices, and bookings.

3. System Administrator

The system administrator manages users, theatres, system configurations, security, and overall system maintenance.

4. Payment Gateway

The payment gateway processes online transactions securely and provides payment status to the booking system.

5. Theatre Management

Theatre management can use booking information and reports to monitor ticket sales, occupancy, and revenue.

2.2 Roles and Responsibilities

Stakeholder	Responsibilities
Customer	Search movies, select shows, book seats, make payments and view tickets
Theatre Administrator	Manage movies, screens, seats, shows and ticket prices
System Administrator	Manage users, security, databases and system maintenance
Payment Gateway	Process and verify online payments
Theatre Management	Monitor bookings, occupancy and revenue

2.3 Major Challenges

The major challenges include:

- Maintaining accurate real-time seat availability.

- Preventing two users from booking the same seat.
- Handling payment failures.
- Protecting customer information.
- Managing a large number of users during popular movie releases.
- Providing high system availability.
- Handling booking cancellation and refunds.
- Maintaining accurate booking records.

3. Analyze the Current Approach

3.1 Existing System

In a traditional movie ticket booking system, customers generally visit the theatre directly to purchase tickets. The customer asks the booking counter about available movies and show timings. The staff checks seat availability and issues the ticket after receiving payment.

Some theatres also provide basic online booking facilities, but the features may be limited depending on the theatre.

3.2 Strengths of Existing System

- Simple process for customers who visit the theatre.
- Direct interaction with theatre staff.
- Cash payment can be supported.
- Immediate physical ticket collection.
- Suitable for customers who are not comfortable with online systems.

3.3 Limitations

- Customers may have to wait in long queues.
- It requires physical presence at the theatre.
- Customers cannot easily compare different show timings.
- Seat selection may be inconvenient.

- Manual processes can result in errors.
- Ticket availability may not be known before visiting the theatre.
- Large crowds can increase waiting time.
- Manual record maintenance is difficult.

3.4 Gaps to be Addressed

The proposed system should address these problems by providing:

- Online access to movie information.
- Real-time seat availability.
- Online seat selection.
- Secure digital payments.
- Electronic tickets.
- Booking history.
- Automated cancellation and refund processing.
- Centralized database management.

4. Proposed Solution Using Software Engineering Principles

4.1 Proposed System

The proposed Online Movie Ticket Booking System is a web-based software application that connects customers, theatres, administrators, and payment services.

The customer logs into the system, searches for a movie, selects a theatre and show, chooses available seats, makes payment, and receives an electronic ticket.

4.2 Major Functional Modules

1. User Management Module

- User registration.
- Login and logout.

- Password management.
- User profile management.

2. Movie Management Module

- Add movies.
- Update movie details.
- Remove movies.
- Display movie title, genre, language and duration.

3. Theatre Management Module

- Manage theatres.
- Manage screens.
- Manage seating arrangements.
- Manage ticket prices.

4. Show Management Module

- Add show timings.
- Update show timings.
- Display available shows.
- Associate movies with particular screens.

5. Seat Booking Module

- Display available seats.
- Allow users to select seats.
- Temporarily lock selected seats.
- Confirm booked seats after successful payment.

6. Payment Module

- Calculate total ticket cost.

- Connect with payment gateway.
- Process online payment.
- Handle successful and failed transactions.

7. Ticket Module

- Generate electronic ticket.
- Provide booking ID.
- Display movie, theatre, show and seat details.
- Allow users to access booking history.

8. Cancellation Module

- Allow cancellation according to theatre rules.
- Update seat availability.
- Initiate refund when applicable.

9. Admin Module

- Manage users.
- Manage movies and theatres.
- Manage shows and seats.
- Monitor bookings.
- Generate reports.

4.3 Functional Requirements

The system shall:

1. Allow users to register and log in.
2. Display available movies.
3. Display theatres and show timings.
4. Display real-time seat availability.

5. Allow users to select seats.
6. Calculate ticket prices.
7. Process online payments.
8. Generate electronic tickets.
9. Store booking history.
10. Allow cancellation of eligible bookings.
11. Allow administrators to manage movies, theatres, shows and bookings.
12. Generate booking and revenue reports.

4.4 Non-Functional Requirements

Performance

The system should respond quickly even when many users access it simultaneously.

Security

User passwords and payment-related information must be protected. Secure authentication and encrypted communication should be used.

Reliability

The system should accurately maintain booking and payment information.

Availability

The system should be available whenever customers need to book tickets.

Usability

The interface should be simple and easy to understand.

Scalability

The system should support an increasing number of users, theatres, movies and bookings.

Maintainability

The software should be modular so that individual components can be modified without affecting the entire system.

4.5 Application of Software Engineering Principles

Modularity

The system is divided into separate modules such as User Management, Movie Management, Booking, Payment and Ticket Generation. This makes development and maintenance easier.

Scalability

The system can use scalable cloud infrastructure and database technologies so that additional users and theatres can be supported.

Maintainability

Clean coding practices, documentation and modular architecture make future maintenance easier.

Reliability

Database transactions and booking validation can ensure that incorrect or duplicate bookings are avoided.

Usability

A simple user interface allows customers to search for movies, select seats and complete bookings easily.

Security

Authentication, authorization, encrypted communication and secure payment gateway integration help protect user information.

5. Justify the Chosen Methodology/Framework

5.1 Selected Methodology – Agile Scrum

The **Agile Scrum methodology** is suitable for developing the Online Movie Ticket Booking System.

Agile development divides the project into small iterations called **Sprints**. Each Sprint delivers a working part of the system.

5.2 Why Agile Scrum is Suitable

The requirements of an online booking system can change based on customer and theatre requirements. Agile allows the development team to respond to these changes quickly.

For example, the first Sprint can develop user registration and login. The next Sprint can implement movie and theatre search. Later Sprints can implement seat booking, payment and ticket generation.

Example Sprint Plan

Sprint	Major Activities
Sprint 1	Requirements, user registration and login
Sprint 2	Movie and theatre management
Sprint 3	Show timings and seat availability
Sprint 4	Seat booking and payment
Sprint 5	Ticket generation and cancellation
Sprint 6	Testing, security and deployment

5.3 Benefits of Agile Scrum

- Continuous customer feedback.
- Early delivery of working software.
- Easy adaptation to changing requirements.
- Continuous testing.
- Faster identification of errors.
- Better communication among development team members.
- Regular improvement of the software.

Agile also supports maintenance because new features and improvements can be added through future Sprints.

6. Expected Outcomes and Limitations

6.1 Expected Outcomes

The proposed system is expected to:

1. Reduce waiting time for customers.
2. Provide convenient online booking.
3. Provide real-time seat availability.
4. Reduce manual booking errors.
5. Provide secure online payments.
6. Generate electronic tickets.
7. Improve theatre management.
8. Provide booking and revenue reports.
9. Improve customer satisfaction.
10. Reduce the workload of theatre staff.

6.2 How the Solution Addresses the Challenges

Challenge	Proposed Solution
Long queues	Online ticket booking
Seat availability uncertainty	Real-time seat display
Duplicate booking	Database transaction and seat locking
Payment problems	Secure payment gateway
Manual records	Centralized database
Customer inconvenience	Web/mobile access

High traffic	Scalable architecture
Data security	Authentication and encryption

6.3 Potential Risks

- Server failure.
- Network problems.
- Payment gateway failure.
- Cybersecurity attacks.
- Database failure.
- High traffic during popular movie releases.
- Incorrect seat synchronization.
- User privacy issues.

6.4 Implementation Challenges

The major implementation challenges include maintaining real-time seat availability, integrating payment gateways, protecting user data, handling high traffic and ensuring reliable database transactions.

6.5 Limitations

- Requires an internet connection.
- Online payment may not always be successful.
- System downtime can temporarily prevent bookings.
- Integration with multiple theatres may require additional development.
- Refund processing may depend on payment providers and theatre policies.

6.6 Future Enhancements

The system can be improved in the future by adding:

- Android and iOS mobile applications.

- AI-based movie recommendations.
- Multiple-language support.
- Digital wallet integration.
- Loyalty and reward programs.
- Dynamic pricing.
- QR-code-based theatre entry.
- Voice-based movie search.
- Personalized offers.
- Integration with food and beverage ordering.

Conclusion

The Online Movie Ticket Booking System provides a convenient and efficient solution for replacing or improving traditional movie ticket booking. It allows customers to search for movies, select theatres and show timings, choose seats, make secure payments and receive electronic tickets.

The proposed system applies important software engineering principles such as modularity, scalability, maintainability, reliability, usability and security. Agile Scrum is selected as the development methodology because it supports incremental development, continuous feedback and changing requirements.

Therefore, the proposed system can improve customer convenience, reduce manual work, minimize booking errors and help theatres manage their operations more effectively.